

## 1 Title Slide

# GOLD PRICE ANALYSIS AND PREDICTION USING MACHINE LEARNING-BASED FORECASTING

Model: Random Forest Regression

- Dataset: Gold Price 2016 to 2026
- Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn
- Goal: Predict Future Gold Prices Using ML



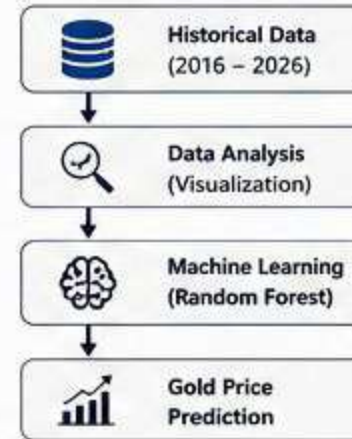
## 2 Introduction

### Problem Statement

Gold price prediction is challenging due to market volatility. Accurate forecasting helps investors and financial institutions in decision making.

### Objective

- Analyze historical data
- Identify key factors
- Build ML model
- Predict future gold price
- Evaluate model performance



## 3 Dataset Description

| Feature | Description            |
|---------|------------------------|
| Date    | Trading Date           |
| Open    | Opening Price          |
| High    | Highest Price          |
| Low     | Lowest Price           |
| Close   | Closing Price (Target) |
| Volume  | Trading Volume         |



Dataset Period  
2016 - 2026

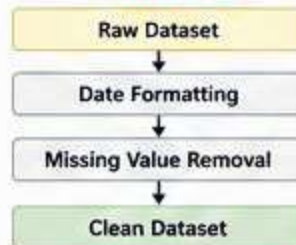


Total Records  
~ 2500+ (Approx.)

## 4 Data Preprocessing

- Converted 'Date' column to datetime format
- Removed missing values
- Checked and cleaned data
- Dataset is ready for model building

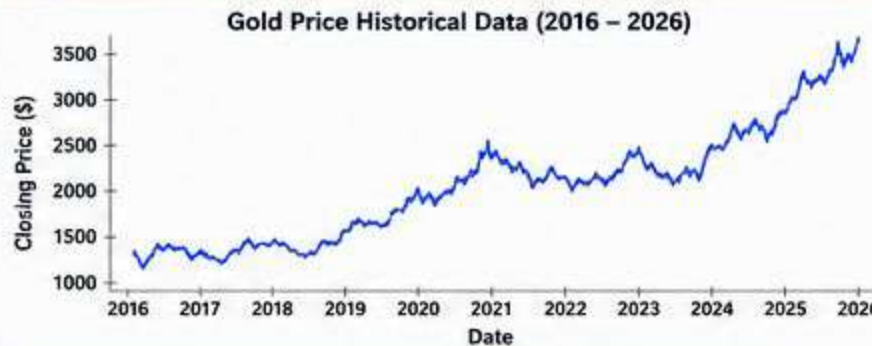
### Process Flow



### Results

| Process        | Result                |
|----------------|-----------------------|
| Missing Values | Removed               |
| Date Format    | Converted to datetime |
| Dataset Status | Clean and Ready       |

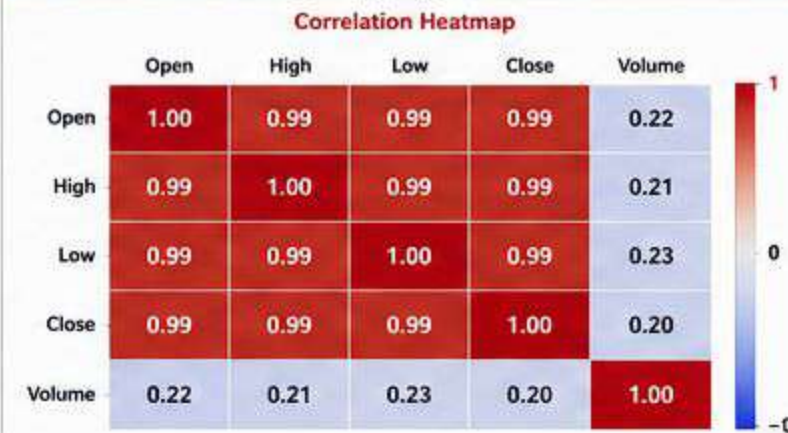
## 5 Data Visualization – Gold Price Trend



### Key Insights

- Gold price remained stable during 2016-2018.
- Significant rise seen after 2019.
- Sharp increase around 2020 (COVID-19 impact).
- Continuous growth trend observed till 2026.
- Indicates strong market demand.

## 6 Correlation Analysis (Heatmap)



### Interpretation

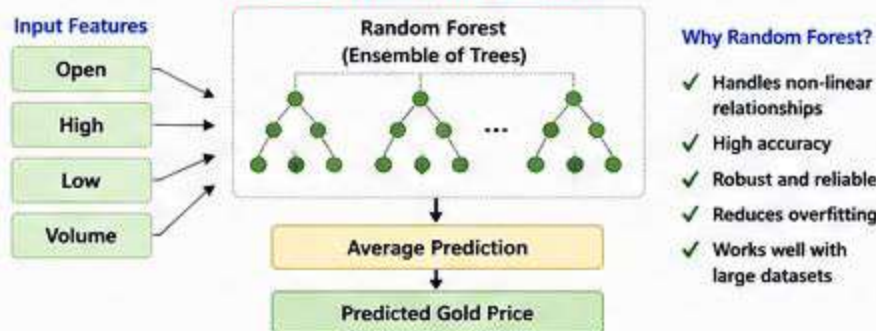
- Open, High, Low and Close are strongly positively correlated.
- Volume has weak correlation with Close price.

### Conclusion

Open, High and Low prices are the most influential features.

## 7 Proposed Methodology

### Random Forest Regression

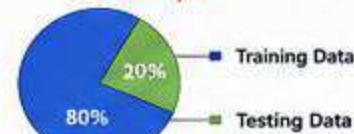


## 8 Model Training and Prediction

### Feature Selection

X = (Open, High, Low, Volume)  
y = Close

### Train-Test Split



### Model Used

```
RandomForestRegressor(  
    n_estimators=100,  
    random_state=42  
)
```

### Training Details

- Training Data : 80%
- Testing Data : 20%
- Random State : 42
- Trees (n\_estimators) : 100

## 9 Results – Actual vs Predicted



### Interpretation

- Predicted prices closely follow actual prices.
- Model captures market trends effectively.
- Very small prediction error.

## 10 Model Performance

### Evaluation Metrics

- R<sup>2</sup> Score – Measures model accuracy
- MSE – Measures average squared error

### Performance Results

| Metric                   | Value   |
|--------------------------|---------|
| R <sup>2</sup> Score     | 0.9878  |
| Mean Squared Error (MSE) | 9156.24 |

### Interpretation

- R<sup>2</sup> Score is close to 1 → Excellent accuracy
- Low MSE indicates minimal prediction error
- Model performs extremely well.

High Accuracy ✓ Low Error ✓ Reliable Model ✓

## 11 Future Gold Price Prediction

### Input Sample Data

| Open | High | Low  | Volume |
|------|------|------|--------|
| 3300 | 3320 | 3280 | 120000 |



### Predicted Future Gold Price

3325.41  
USD

Note: The predicted price will vary with different input values.

## 12 Conclusion

### Key Takeaways

- Historical gold prices show strong upward trend.
- Open, High and Low prices have high impact on closing price.
- Random Forest Regression provides high prediction accuracy.
- Actual vs Predicted results show strong alignment.
- Model can be used for future gold price forecasting and investment decision support.



Random Forest Regression is an effective and reliable model for Gold Price Prediction.